

FREE EXPERIENCE CLASS · 体验课

AGENT ARCADE

Teach the Bot. Test the Bot. Break the Bot.

教会它 · 测试它 · 找出它的弱点

Thursday, August 20, 2026 · 60 minutes · Grades 8–12 · no installation

▶ Launch Agent Arcade

Open mission sheet

COLD OPEN · NO EXPLANATION YET

Who will survive an unseen level: you, your bot, or neither?



Vote first: What will make the biggest difference—more rows, more diverse levels, or a more complicated model?

TODAY'S MISSION · 你要完成什么

One game. One agent. One defensible result.



You leave with

Your gameplay CSV, a trained browser agent, a controlled-experiment receipt, and a one-paragraph research claim.

No setup tax

No Python, API key, account, download, camera, or microphone.
Everything runs locally in one browser tab.

SCOPE · EXCITING AND SAFE

This is an AI game-agent sandbox—not a commercial-game cheat.

Inside our sandbox

- We own the game rules and test levels.
- The agent sees only four numeric sensors.
- We can pause, inspect, and reproduce every decision.

Outside today's scope

- No connection to Fortnite, Roblox, Minecraft, or other services.
- No bypassing anti-cheat or access controls.
- No collection of identity or private behavior.

Research name: **automated game-playing agent** · Classroom nickname: “AI外挂实验室”

GAME RULES · 90-SECOND BRIEFING

Collect coins. Avoid hazards. Every key press becomes a label.



Choose a lane

Arrow keys or A / D. On a phone, tap the left or right half.



Collect

Align with a coin before it reaches the player.



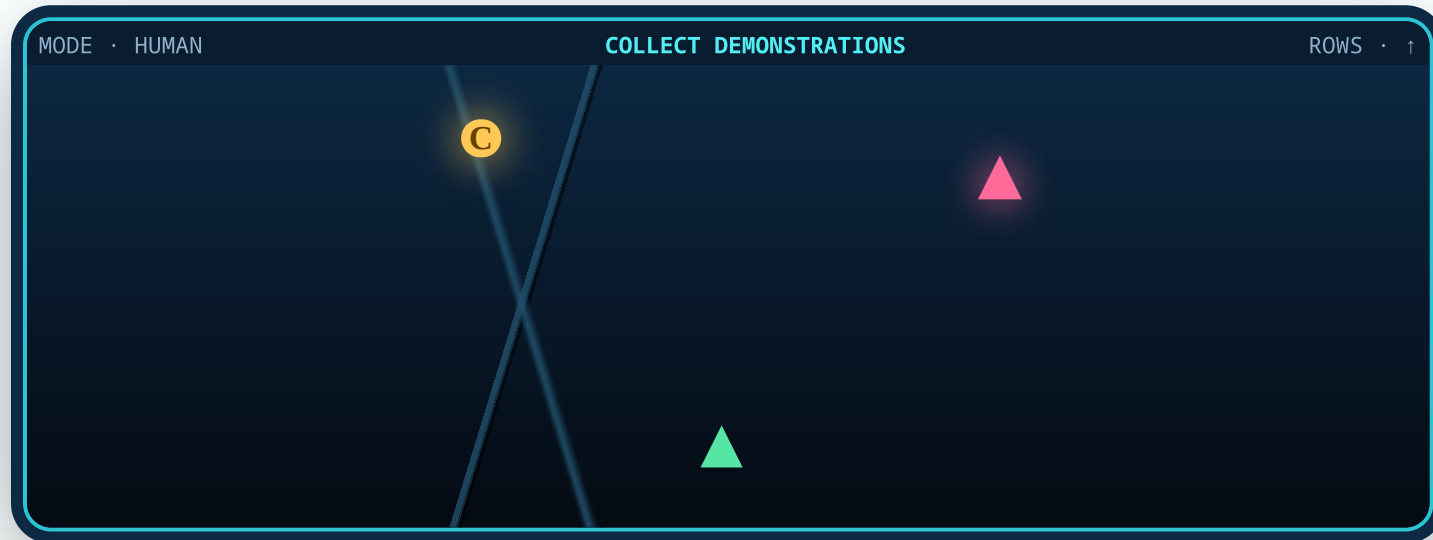
Avoid

Three collisions end the current episode.

[▶ Open the game in a new tab](#)

LAB 1 · 8 MINUTES · TEACH THE BOT

Play two rounds—and become the dataset.

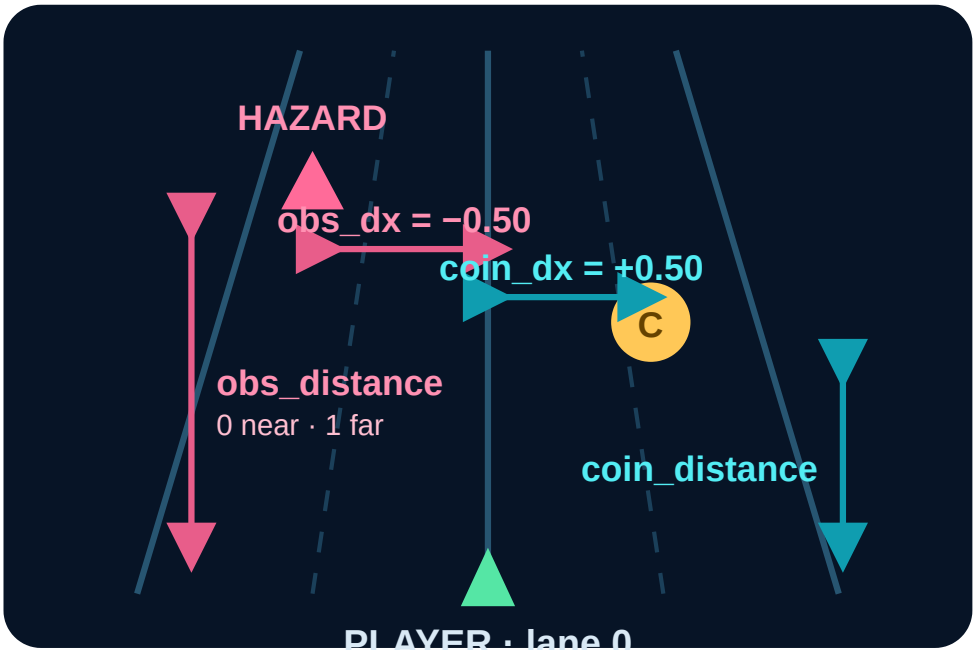


Your checklist

1. Press **Player Run**.
2. Play until you have ≥ 80 personal examples.
3. Try both left and right moves.
4. Record one failure in Part A.

Notice: doing nothing is also an action: STAY.

One frame becomes four measurements, one action, and one outcome.



Field	Plain-English meaning
obs_dx	obstacle horizontal offset: - left · 0 same lane · + right
obs_distance	forward distance to nearest hazard: 0 near · 1 far / none
coin_dx	coin horizontal offset: - left · 0 same lane · + right
coin_distance	forward distance to nearest coin: 0 near · 1 far / none

dx means “delta x”: horizontal position relative to the player—not absolute screen pixels.

STATE · what it sees

`[-0.50, 0.18, +0.50, 0.44]`

ACTION · your label

RIGHT

REWARD · what followed

+0.02 survival

QUICK CHECK · WHO IS DOING THE LABELING?

Quick check · tap an answer

During Player Run, what is the training label for each row?

The level seed

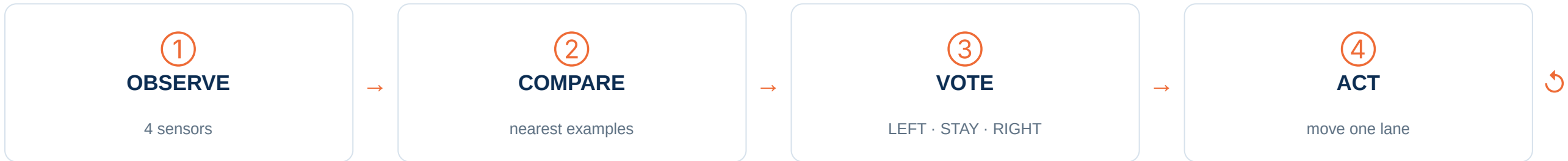
The player's LEFT / STAY / RIGHT action

The coin color

The final score only

THE AGENT LOOP · EVERY 100 MILLISECONDS

Observe → **compare** → **choose** → **act** → **observe again**



Policy: a function that maps a state to an action. The game's AI is an agent because its choices change the next state it will observe.

THE MODEL · SIMPLE ENOUGH TO INSPECT

Weighted k-NN asks: “What did players do in similar situations?”

Current state

[obstacle Δ , obstacle distance,
coin Δ , coin distance]



Find the 9 closest demonstration rows.

Weighted vote

LEFT 18% · STAY 27% · **RIGHT 55%**



The closest examples receive more weight.

A more complicated model is not automatically a better first experiment.

Nine lines turn nearby demonstrations into the next action.

```
1 function predict(state, rows, k = 9) {  
2   const x = vector(state);  
3   const near = rows.map(r => ({ action: r.action,  
4     d: r.x.reduce((sum, v, i) => sum + (v - x[i]) ** 2, 0)  
5   })).sort((a, b) => a.d - b.d).slice(0, k);  
6   const vote = { "-1": 0, "0": 0, "1": 0 };  
7   near.forEach(n => vote[n.action] += 1 / (.025 + Math.sqrt(n.d)));  
8   return [-1, 0, 1].sort((a, b) => vote[b] - vote[a])[0];  
9 }
```



Interactive JavaScript walkthrough

Press **Explain each line** to hear the walkthrough and watch the matching code lines highlight.

READ CODE LIKE A SCIENTIST · LINE BY LINE

Every line makes a claim we can test.

- 01 `Contract:` one state + demonstration rows → one action.
- 02 `Represent:` convert the visible situation to four numbers.
- 03–05 `Compare:` square feature differences, sort by distance, keep the nearest nine.
- 06 `Initialize:` start LEFT, STAY, and RIGHT votes at zero.
- 07 `Weight:` closer examples contribute a larger vote; `.025` prevents division by zero.
- 08 `Decide:` sort actions by vote and return the winner.

Debugging question: If every demonstration says STAY, which line is wrong? **None.** The data—not the syntax—fails to cover LEFT and RIGHT.

Train your agent, then hand over the controls.

1 · Train

Press **Train My Agent**. Read the training receipt: rows and action balance.

2 · Take over

Press **AI Takeover**. Keep your hands off the keyboard and watch its sensor rays.

3 · Break it

Press **Unseen Level**. Record one success and one failure in Part B.

Ask while watching: Is this a bad decision, a missing example, an insufficient sensor, or simply a hard situation?

[Return to your running game](#)

READ THE OVERLAY · DO NOT ANTHROPOMORPHIZE

The animation is not the explanation—the decision receipt is.



Three layers of explanation

1. **Observation:** exact sensor values.
2. **Decision:** action probabilities.
3. **Outcome:** reward and collision.

“The AI wanted the coin” is a story. “RIGHT had the highest weighted vote” is inspectable.

FROM GAME TO RESEARCH · TURN FUN INTO A TEST

Our question changes one thing—and fixes the data budget.

With the same 240 demonstration rows, does training on more level seeds improve action accuracy on unseen levels?

Change

1 seed versus **12 seeds** in training demonstrations.

Hold fixed

240 rows, four sensors, weighted k-NN, k=9, and the same test seeds.

Measure

Accuracy on 600 expert-labeled actions from 20 unseen seeds; game score as a secondary outcome.

Quick check · tap an answer

The 12-seed agent has 2,400 training rows; the 1-seed agent has 240. Can a score difference be attributed to level diversity?

Yes—more seeds always means diversity

No—training amount also changed

Yes—both use k-NN

Only if the score is above 90%

Same rows. Same model. New test levels.

In Agent Arcade

1. Scroll to **Generalization test**.
2. Press **Run fair test**.
3. Record both accuracies and mean scores.
4. Calculate the percentage-point difference.

Before seeing the result

Write your prediction and mechanism:

“I predict ____ because diverse levels should / should not ____.”

▶ Run the generalization test

WRITE THE RESULT · OBSERVATION BEFORE EXPLANATION

Use a four-part evidence sentence.

①

SCOPE

this browser experiment

+

②

RESULT

two measured accuracies

+

③

LIMIT

one game · one model

+

④

NEXT TEST

repeat and extend

Sentence frame ▶ In this three-lane game, the agent trained on [12 / 1] seed(s) achieved [A]% versus [B]% unseen-level action accuracy with 240 rows each. Because we tested one environment and one k-NN setting, the result is preliminary. Next, we would repeat across seeds and compare another policy model.

OPEN RESEARCH BACKING · THIS QUESTION ALREADY MATTERS

Our tiny game mirrors real benchmarks—at classroom scale.

Procgén

16 procedurally generated game-like environments designed to measure sample efficiency and generalization.

[Paper](#) · [MIT-licensed code](#)

GVGAI

An open framework for agents that must play many games, including games they did not train on.

[Official repository](#)

NetHack Learning Dataset

Large collections of human and symbolic-bot trajectories support imitation and human-versus-bot research.

[NeurIPS dataset paper](#)

OPEN-DATA EXTENSION · REAL PLAYERS, LICENSED SOURCE

Can four interpretable sensors retain most of a full telemetry model?

SkillCraft1 · UCI

3,395 StarCraft II player records · **15** behavioral telemetry fields · target is $\text{LeagueIndex} \geq 5$.

[Dataset DOI](#) · Blair, Thompson, Henrey & Chen (2013) · CC BY 4.0

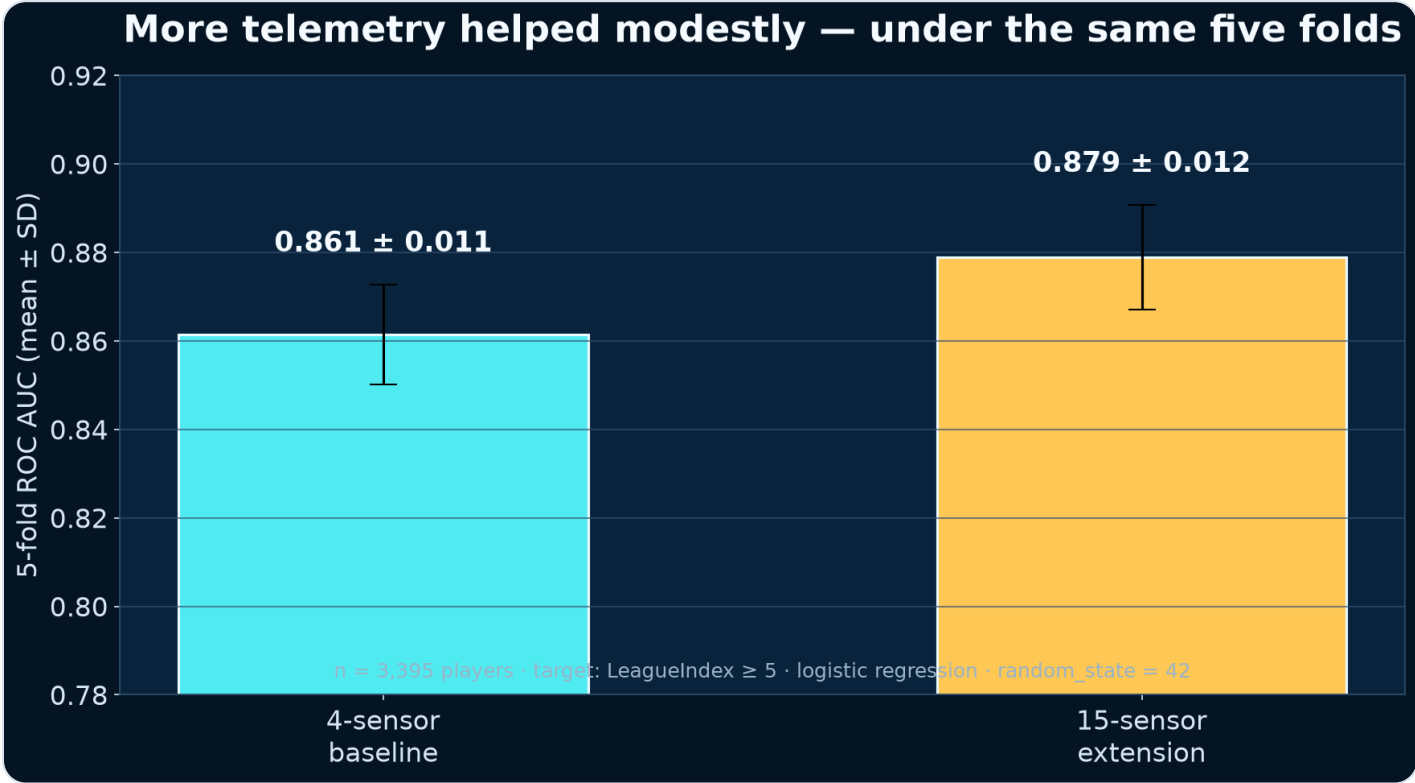
Controlled comparison

4 sensors: APM, hotkey selection, action latency, perception–action cycles.

15 sensors: all behavioral telemetry. Same players, target, model, preprocessing, and five folds.

RQ: How much predictive performance is lost when we use four interpretable sensors instead of all fifteen?

The compact sensor set retained most—but not all—of the signal.



4 sensors

ROC AUC 0.861 ± 0.011
accuracy 77.6% ± 1.3%

15 sensors

ROC AUC 0.879 ± 0.012
accuracy 78.9% ± 1.2%

Mean ± SD across five stratified folds; class-balanced logistic regression; random_state=42. Observed ROC AUC difference: +0.018. [Open evidence package](#)

[OPEN DATA](#) → [EXTENDED ABSTRACT](#) · [EXACT AI HANDOFF](#)

The prompt carries the evidence boundary into the writing.

Prompt used ▶ Use only this evidence receipt: $n=3,395$; same 5 folds; 4-sensor AUC 0.861 ± 0.011 ; 15-sensor AUC 0.879 ± 0.012 ; observed $\Delta=+0.018$. Draft Background, Method, Results, Limitations, and Next Experiment. Say “associated with,” not “caused.” Do not invent p-values or confidence intervals. State that LeagueIndex is an operational label.

[See the full exact prompt, generated draft, source hash, code, fold results, and disclosure.](#)

Generated result

Title: How Much Telemetry Is Enough?

Bounded takeaway: four sensors retained most of the discrimination; all 15 produced a modest gain that motivates a sensor-ablation study.

[Read the extended abstract](#)

Ask AI to challenge the experiment—not invent the result.

You ▶ Act as a skeptical game-AI researcher. Here is my experiment receipt: 240 rows per condition; 1 versus 12 training seeds; weighted k-NN with k=9; 600 held-out actions from 20 unseen seeds; observed accuracies [A] and [B]. Separate (1) observation, (2) possible explanations, (3) confounders, and (4) the next controlled experiment. Do not invent missing results or claim causation.

Good AI work

Generate competing explanations, check clarity, propose an ablation, draft a bounded caption.

Still your responsibility

Supply exact numbers, verify the code path, reject unsupported claims, disclose AI assistance.

THE TRANSFERABLE IDEA · ONE PIPELINE, MANY DOMAINS

The game is our sandbox. The research pipeline travels much farther.



What transfers: define the evidence, test fairly, inspect failures, and improve the next experiment.

EXTENSION PATH · FROM CLASS ARTIFACT TO PUBLICATION

The same project can grow without changing its central question.



Venue fit: IEEE Conference on Games · AIIDE · Foundations of Digital Games. A venue is a research community, not a promise of acceptance.

2027 TARGET CYCLE · USE FUTURE DEADLINES

Build toward one confirmed deadline—and label unknown dates honestly.

IEEE Conference on Games 2027 · full paper	01 MAR 2027 · AoE
IEEE CoG 2027 · auxiliary / short paper	15 MAY 2027 · AoE
AIIDE 2027 · official CFP	TBD
FDG 2027 · official CFP	TBD

Our readiness deadlines: question freeze · 15 Oct 2026 → preregistered experiment · 15 Dec → full results · 31 Jan 2027 → CoG manuscript · 22 Feb.

Confirmed CoG dates: [official CoG 2027 site](#). AIIDE and FDG dates are shown as TBD until their official 2027 CFPs are released.

MAKE THE ARTIFACT · 6 MINUTES

Finish your one-page Agent Research Card.

1. **Question:** what changed, what stayed fixed?
2. **Data:** what does one row represent?
3. **Result:** counts/accuracies plus the difference.
4. **Failure:** one concrete agent mistake.
5. **Boundary:** what can't this test establish?
6. **Next experiment:** one change, one outcome.

Peer check

Can I reproduce the numbers?

Is the comparison fair?

Does the claim stay inside the evidence?

Could I run the proposed next test?

[Open / print card](#)

EXIT TICKET · 最后 60 秒

THE BOT PLAYED. YOU DID THE RESEARCH.

One thing I taught

A state → action pattern through demonstrations.

One thing I tested

Generalization under a fixed data budget.

One thing I still doubt

A failure, confounder, or untested population.

[Replay Agent Arcade](#)

[Explore the 15-class course](#)

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